Math Syllabus

Grades 1–4

2026–2027

INSTRUCTOR INFORMATION

Instructor	Email	Class Days
Tiffany Young	Tiffany@rivertech.me	Monday–Friday
Mary Hegelund	Mary@rivertech.me	Monday–Friday

COURSE OVERVIEW

Math develops students' number sense, computation skills, problem-solving ability, and mathematical reasoning through structured instruction, hands-on practice, and individualized support.

Students use IXL Ultimate Math Workbooks at their appropriate grade and skill level as the primary curriculum. The workbooks follow a structured, grade-by-grade progression that builds foundational number sense before moving into more advanced computation, fractions, measurement, geometry, and problem solving.

Instruction is reinforced with a variety of manipulatives, including number lines, ten-frames and twenty-frames, counting beads, place-value materials, clocks, rulers, two- and three-dimensional shapes, and fraction models, giving students concrete, hands-on ways to understand abstract math concepts.

Numberblocks videos are used to provide additional visual reinforcement of number sense and early math concepts, and Khan Academy is available for additional practice and support.

Instruction is differentiated so that each student can work at an appropriate level of challenge while building steadily toward grade-level proficiency.

The overall goal is to develop confident, capable mathematicians who understand number relationships, compute accurately, and can reason through and solve problems, while progressing according to their individual level and readiness.

COURSE MATERIALS

Required Materials

- IXL Ultimate Math Workbooks (at the student's appropriate grade and skill level)
- Manipulatives: number lines, ten-frames and twenty-frames, counting beads, place-value materials, clocks, rulers, 2D/3D shapes, and fraction models
- Access to Numberblocks videos for visual reinforcement
- Access to Khan Academy for additional practice and support

SYLLABUS BY SKILL AREA

Number Sense & Place Value

Grade Band	Focus
Grade 1	Counting, comparing and ordering numbers, number lines, number patterns, tens and ones, and beginning place value.
Grade 2	Place value to larger numbers, expanded form, comparing numbers, skip counting, and using place value to support computation.
Grade 3	Place value, rounding, estimation, number patterns, and increasingly large whole numbers.
Grade 4	Place value with large whole numbers, rounding and estimation, factors and multiples, and prime and composite numbers.

Addition & Subtraction

Grade Band	Focus
Grade 1	Modeling addition and subtraction, number lines, making 10, addition and subtraction within 20, equations, and simple word problems.
Grade 2	Addition and subtraction strategies, fact fluency, place-value strategies, regrouping, multi-digit computation, and word problems.
Grades 3–4	Increasingly fluent multi-digit addition and subtraction, estimation, problem solving, and application of operations in more complex situations.

Multiplication & Division

Multiplication and division are introduced progressively across the grades. Grade 2 builds the foundation through grouping, skip-counting, arrays, and repeated addition.

Grade 3 introduces multiplication and division facts directly, using arrays, properties of operations, fact families, and word problems to build fluency within grade-level expectations.

Grade 4 develops more advanced multiplication and division, including multi-digit multiplication, division by one-digit numbers, estimation, and factors and multiples, applied through increasingly complex word problems.

Fractions & Decimals

Grade Band	Focus
Grades 1–2	Equal shares, halves, thirds, fourths, and beginning fraction language through visual and hands-on experiences.
Grade 3	Identifying, representing, comparing, and understanding simple equivalent fractions using models and number lines.
Grade 4	Equivalent fractions, comparing fractions, working with fractions in practical contexts, and beginning decimal concepts, including decimal place value and relationships between fractions and decimals.

Measurement, Time, Money & Data

Students develop practical measurement and data skills across the grades, including measuring length, weight, and capacity using standard and nonstandard units; telling time and working with elapsed time; counting and making change with money; and reading, interpreting, and creating simple graphs and charts.

Instruction becomes increasingly precise and complex across the grade levels, moving from basic measurement and time-telling toward more advanced applications involving multiple units, longer time intervals, and more detailed data sets.

Geometry

Students build understanding of shapes and spatial reasoning progressively, beginning with identifying and describing basic two- and three-dimensional shapes and their attributes, and advancing toward classifying shapes by properties, understanding symmetry, and working with area and perimeter as students progress through the grades.

Mathematical Reasoning & Problem Solving

Problem solving is woven throughout every skill area rather than taught in isolation. Students practice interpreting word problems, choosing appropriate strategies and operations, explaining their reasoning, and checking their work for accuracy.

As students advance, problems increase in complexity, requiring multiple steps, careful reading, and flexible thinking to apply the right combination of skills.

Hands-On Learning & Reinforcement

Manipulatives such as number lines, ten- and twenty-frames, counting beads, place-value materials, clocks, rulers, shapes, and fraction models are used throughout instruction to make abstract concepts concrete.

Numberblocks videos provide additional visual reinforcement of number relationships, and Khan Academy offers further practice opportunities to reinforce and extend skills introduced in the IXL workbooks.

Individualized Learning

Students' math abilities do not always correspond exactly to their grade placement. Instruction is differentiated according to each student's individual skill level and readiness.

Students may work at, above, or below their enrolled grade level in particular math materials. Teachers adjust workbook pacing, manipulative use, and one-on-one support to provide an appropriate level of challenge while building toward grade-level proficiency.

COURSE GOALS

Strong number sense and computational fluency are key goals of the elementary Math program. Our goal is for students to build a solid foundation in number sense, the four operations, fractions, measurement, and geometry by the end of Grade 4. Many students achieve this earlier, while others may need additional time and support. Instruction is adjusted to each student's individual development while continually working toward these goals.

By the end of the course, students will demonstrate continued growth in number sense, place value, addition, subtraction, multiplication, division, fractions, decimals, measurement, geometry, and mathematical reasoning.

Students will increasingly be able to compute accurately, reason through multi-step problems, and apply math skills confidently to real-world situations.